

TASK 2: Define the Scenario and System

TIVIDY is a generic workflow management system which controls the flow of processes from start to finish. Clients specify workflows suited to their needs, manage them and execute them as instances using the system. Due to the generic nature of TIVIDY, clients are not limited to a single workflow but may define multiple workflows which differ in structure, approaches, routing and overall execution. The workflow management system is not responsible for carrying out the actual work (logic) behind the workflow; it is only used to define the sequence of activities that must take place, how they must be completed and who is responsible for what.

Using the workflow manager, clients define the scope of their workflows (workflow definition), which includes activities and their ordering, routing and decision conditions, participants and roles, rules regarding approval and rejections, state transitions, dependencies and constraints. The workflow definition is generic and contains no specific data such as participant names. Workflow instances are created by TIVIDY using the workflow definition. Instances contain meaningful data such as the current state of the workflow, assigned participants, specific work items, decisions that have been made, history of the instance, as well as information specific to the instance. Multiple instances may be created from the same definition. Within the system instances are kept independent and thus changes to one instance will not affect another.

Workflow scenario: Project approval

The chosen scenario used to demonstrate the application of TIVIDY is a project approval workflow in a technology company. The technology company is responsible for creating software solutions as products and services. Before the start of a project the company receives detailed proposals from product teams who communicate with clients. It is important that these proposals are technically and financially feasible thus they must undergo a series of approvals, reviews and changes before execution. The technology company is large-scale and thus multiple proposals are handled simultaneously.

The process being managed is the lifecycle of the project proposal from the time it is submitted to the time it is approved/ rejected. The workflow includes four major participants. The product manager - reviews the proposal submitted by product teams for correctness and completeness. The head software engineer - determines if the project is technically feasible. The head of finance - decides whether the project is profitable and financially feasible. The chief operation officer - provides the final decision of whether the proposal is approved or rejected. Lower level employees may perform work items such as comparing profitability of previous projects to assist major participants in decision making.

Work items may include, submitting a proposal, adding priority to a proposal, review of correctness and completeness, budget comparison, financial feasibility review, risk and security evaluation, technical review, creation of summary report, final review, and beginning project.

Routing is a major part of this workflow. In terms of control flow routing sequential and conditional routing is utilized. The proposal must be reviewed by at most one participant at a time and may only be passed on to the next participant if approved, otherwise rejection/ change requests would redirect the proposal to previously visited participants. On the other hand, in terms of resource routing, role-based routing has been chosen . Work items are assigned to participants based on their job title.

Participants are responsible for approvals. Whether or not the proposal is passed on to the next participant is dependent on the decision of the participant currently holding the proposal. For example the technical review is dependent on the approval of the completeness review. The financial review is dependent on the technical review approval etc.

The proposal can undergo multiple failures/ exceptions during the workflow instance. The proposal may be incomplete and rejected during the completeness review by the product manager. It may be decided that the project exceeds the budget and is rejected by the head of finance. Due to prioritization of other proposals the approval deadline of the proposal may be missed.

The workflow may have two outcomes. In the best case the proposal is approved by all departments and reaches the state “approved” and is therefore given the greenlight to proceed with implementation. In the second outcome the proposal may be rejected by one of the departments, even after reviews and changes it may be deemed as “rejected” and will not be revisited.